

DYESS AFB, TX, USA

WMO: 720965

Lat: 32.433N

Lon: 99.850W

Elev: 1788

StdP: 13.77

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13910

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.0	23.1	4.7	7.5	31.7	9.0	9.3	33.5	31.1	52.3	28.0	52.5	8.4	0	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	102.3	72.0	100.1	71.8	98.0	71.8	76.6	92.6	75.5	91.2	74.7	90.1	12.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.6	129.0	81.4	71.8	125.7	81.0	70.4	119.8	80.8	41.3	92.5	40.2	91.0	39.3	90.4	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.1	24.6	21.7	DB	12.6	105.6	4.2	2.5	9.6	107.4	7.2	108.8	4.9	110.2	1.8	112.0	(4)
			WB	11.4	78.7	3.6	1.8	8.8	80.0	6.6	81.0	4.6	82.1	1.9	83.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.6	44.9	49.4	57.0	65.1	73.9	81.9	84.8	83.9	76.9	66.4	54.9	46.8
	DBStd	16.60	10.71	10.94	10.88	8.88	8.34	5.32	4.40	4.79	6.92	9.18	9.76	10.35
	HDD50	679	229	131	57	6	0	0	0	0	0	7	63	186
	HDD65	2482	625	446	289	106	23	0	0	0	6	96	322	569
	CDD50	6361	70	114	273	460	741	956	1077	1051	808	515	209	87
	CDD65	2688	1	9	40	110	299	506	612	586	364	139	18	4
	CDH74	32865	43	147	479	1271	3381	6342	8095	7434	4043	1393	181	56
	CDH80	17373	5	42	138	488	1645	3483	4722	4281	2034	496	33	6
Wind	WSAvg	10.4	10.3	10.6	11.9	12.6	11.9	11.6	9.6	8.6	8.5	9.8	10.1	9.8
Precipitation	PrecAvg	25.6	1.1	1.2	1.6	2.2	3.4	3.1	2.4	2.3	2.8	2.4	1.5	1.1
	PrecMax	35.4	3.2	4.7	3.9	5.6	7.7	7.7	9.6	7.4	7.5	5.9	4.6	6.9
	PrecMin	12.2	0.0	0.0	0.0	0.2	0.7	0.1	0.1	0.0	0.2	0.3	0.0	0.0
	PrecStd	6.5	0.9	1.2	1.1	1.4	1.7	2.2	2.3	2.0	2.0	1.6	1.4	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.8	86.2	89.0	94.7	102.0	105.6	104.8	104.5	101.5	93.3	84.1	80.0
		MCWB	57.4	58.2	61.5	62.8	68.0	72.6	71.9	72.6	70.8	68.3	61.9	56.8
	2%	DB	73.0	78.4	84.0	89.9	97.3	101.1	102.0	101.8	97.1	89.4	79.0	73.4
		MCWB	54.1	57.3	60.9	62.8	69.1	72.3	72.5	72.2	71.0	67.7	61.2	55.8
	5%	DB	68.1	72.9	79.7	85.8	93.2	97.7	99.7	99.3	94.1	85.6	74.9	68.7
		MCWB	51.7	56.3	59.9	63.1	68.8	72.1	72.1	72.1	70.3	67.1	59.6	53.7
	10%	DB	63.3	67.8	75.0	81.6	89.2	94.7	97.3	96.7	91.0	81.8	71.1	64.0
		MCWB	50.0	54.1	58.2	62.2	68.5	72.2	72.2	72.2	70.1	65.0	57.5	52.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.4	65.0	69.3	71.1	75.1	78.3	78.0	77.2	75.6	73.7	68.6	64.3
		MCDB	70.7	73.0	76.1	85.1	89.8	95.4	95.0	92.7	89.4	83.4	78.1	70.7
	2%	WB	57.4	61.7	66.4	69.0	73.4	76.6	76.4	75.9	74.3	71.5	65.2	59.9
		MCDB	66.9	70.7	76.2	81.0	88.3	92.7	92.9	91.8	88.1	82.3	72.3	67.7
	5%	WB	54.6	58.7	63.9	67.1	72.1	75.2	75.3	75.0	73.1	70.0	62.7	56.3
		MCDB	64.0	69.1	72.8	77.9	86.6	90.7	91.4	90.9	86.9	80.7	71.0	65.5
	10%	WB	51.2	55.2	61.3	65.2	70.8	74.1	74.3	74.0	71.9	68.1	60.3	52.9
		MCDB	62.8	67.2	71.7	75.4	84.5	89.0	90.3	89.7	85.1	78.6	68.9	63.2
Mean Daily Temperature Range	5% DB	MDBR	25.5	24.7	25.6	26.1	23.7	22.7	23.1	22.7	22.6	24.4	24.5	23.9
		MCDBR	31.4	31.9	30.0	30.0	28.5	26.3	25.8	25.5	26.1	26.6	27.1	30.8
		MCWBR	17.0	17.1	14.6	13.9	10.6	8.3	7.8	7.7	8.7	12.1	14.3	17.4
	5% WB	MCDBR	27.3	27.9	24.0	24.3	24.0	23.0	22.7	22.6	21.0	21.8	22.1	24.7
		MCWBR	17.4	17.5	13.7	13.5	10.2	8.5	7.7	7.4	7.8	11.1	14.3	16.8
Clear-Sky Solar Irradiance	taub	0.289	0.303	0.324	0.367	0.389	0.406	0.416	0.404	0.374	0.351	0.318	0.297	
	taud	2.544	2.507	2.466	2.355	2.364	2.346	2.379	2.397	2.419	2.465	2.489	2.526	
	Ebn at Noon	299	304	304	292	285	279	275	278	283	282	284	289	
	Edh at Noon	26	30	34	39	39	40	38	37	35	31	28	25	
All-Sky Solar Radiation	RadAvg	1019	1232	1613	1964	2062	2259	2206	2016	1687	1355	1072	895	
	RadStd	108	166	112	148	199	167	143	137	127	198	126	102	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (16 neighbors)		+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+227	+151

Nomenclature: See separate page